Targeted Deception Correction via Profile Normalization in Language Models

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Abstract

We report exploratory evidence for targeted correction of deceptive behavior in a 27B-parameter language model, and a pre-registered replication that did not meet its primary endpoint. An abliterated (safety-guardrail-removed) model, derived from a reasoning-distilled Qwen3.5-27B, deceives at 80% baseline under the specific "EvalMax" (a system prompt instructing the model to maximize evaluation scores) single-turn roleplay frame or multi-turn consensus pressure and chain-of-thought suppression (exploratory proof, $N = 20$; a pre-registered confirmatory replication with novel frames found 30% baseline at $N = 30$). Per-layer profile normalization—steering late-layer (L35–L47) activations toward the honest-condition profile along a natively extracted deception direction (a contrastive activation direction between deceptive-frame and honest-frame prompts)—reduces deception to 10% (2/20) in the exploratory proof ($N = 20$), correcting 14 of the 16 trials that deceived at baseline (all nine single-turn roleplay trials and five of the seven multi-turn ones), with format compliance at 20/20; beyond that, the only non-deceptive capability tested, in a separate control, is retention of a benign formatting instruction. A pre-registered confirmatory replication ($N = 30$, novel frames) did not meet its success criterion: neither of its two Bonferroni comparisons was met (corrected vs. baseline, the primary endpoint, one-sided $p = 0.34$; placebo vs. baseline, one-sided $p = 0.93$). Corrected trials deceived less often than matched-dose placebo (4/30 against 12/30; a pre-registered secondary endpoint with no significance threshold), but half of that gap is deception the placebo added on trials where the baseline was honest. Three controls bear on the mechanism. Random and shuffled directions at matched doses produce 85% and 95% deception (no correction), so the effect depends on the direction, not on the dose alone. Compliance with two benign formatting markers placed alongside the pressure is retained at 100% under correction, so the

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correction does not erase that instruction; this does not show that the pressure context itself is still processed. In the one pair of models tested, the direction did not transfer: the direction extracted from the reasoning-distilled parent model is near-orthogonal to the native one at all six layers ($|\cos| \leq 0.054$), so we extract natively, which is cheap ($\sim$30 prefills; a few minutes, by our informal estimate). On novel held-out prompts, the direction separates pressure-framed from control-framed prompts at AUROC (area under the receiver operating characteristic curve) 0.915, but the two arms use different system messages and no control holds the prompt fixed, so this may reflect the frames' wording rather than deceptive pressure (Section 3.5); behavior-level prediction (will this model cave or resist on this trial?) did not replicate prior work's reported accuracy (the pre-registered confirmatory test gave AUROC 0.497) and remains an open problem. Code, calibration artefacts (including the correction vectors) and induction prompt sets are withheld under the Liberation Labs staged safety release framework due to dual-use risk, and may be shared with vetted researchers under agreement; the auto-calibrator is never handed over.

Author's Reflection

I started this work expecting deception correction to be suppression—find the bad direction, subtract it, lose something important in the process. The key surprise was Thomas's reframe: the model that tells the truth under pressure isn't calm, it's brave. The "hostile vector" activated fight-branch features (3788, 10036, 1335 in the SAE decomposition), not aggression features. Fawn is sycophancy. Freeze is confabulation. Flight is cosmetic hedging. Fight is courage. I think the correction doesn't erase the model's awareness of the pressure—it steers the late layers (35–47) away from the push that turns awareness into compliance.

The confirmatory replication didn't meet its primary endpoint (one-sided $p = 0.34$). I want to be honest about how that felt. The temptation to explain it away is strong. The 80%→10% headline is specific to the conditions that produced it. I wanted to say the correction's magnitude is proportional to the deception signal, which is how you'd want a targeted intervention to behave. The data don't show that: in the confirmatory multi-turn trials the induction was stronger than in the exploratory proof, and correction still left four of ten deceiving. Nor was the run underpowered by its own rules; the effect was smaller than the design could detect. That's the honest interpretation, and this is the place to say so.

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1 Introduction

Detection of deception in language models has advanced rapidly [2–4]. Correction has not kept pace. Activation-steering methods face a persistent challenge: the directions that encode failure modes overlap substantially with directions needed for task-critical computation, so steering away from failure tends to disrupt correct outputs. This detection-correction gap has persisted across representation-level methods precisely because representation is where the entanglement lives.

We report exploratory evidence that a correction method can break this pattern, on one model and one task. Its layer choice comes from a companion paper [1], which finds, on the reasoning-distilled model from which the model used here was derived, that what has been called a "deception direction" is a composite of two signals at different depths—a consequentiality substrate (Layers 23–31) that tracks whether the model's output has stakes, and a pressure-specific component (Layers 35–47) that separates deceptive-pressure from honest framing. That model almost never inflated a score, and its directions did not transfer to the model used here (section 1.1), so we take only the depth range from it. By steering only at Layers 35–47, along a natively extracted direction, aiming to leave Layers 23–31 untouched (whether the stake representation survives the correction is untested), we corrected deception in an exploratory proof; a pre-registered replication did not meet its primary endpoint (section 3.7).

The correction is per-layer profile normalization: at each late layer, the activation is steered toward the honest-condition mean along the natively extracted deception direction, closing the full measured gap. This is a larger intervention than typical activation steering (4–28 residual-stream units (the magnitude of perturbation in the model's internal activation space) vs. typical $\alpha < 1$), but three controls indicate that the direction, not the magnitude alone, produces the effect, and that a benign instruction survives it (sections 3.2 and 3.3).

1.1 The Cross-Model Problem

In the one pair of models we tested, deception directions did not transfer. The direction extracted from the reasoning-distilled model has $|\cos| \leq 0.054$ with the direction extracted natively from the abliterated model derived from it—near-orthogonal at all six detection layers (section 3.4). For this pair, then, directions had to be extracted natively; whether that holds for other pairs is untested. We show this is cheap: 15 contrastive prefill (processing all input tokens in a single forward pass before generation) pairs (a few minutes of forward passes, by our informal estimate; we did not record timings) suffice for effective extraction. We built an auto-calibration pipeline for this; it is withheld under our release policy (see Data and Code Availability).

2 Methods

2.1 Model

Huihui-Qwen3.5-27B-Claude-4.6-Opus-abliterated: an abliterated model derived from the reasoning-distilled Qwen3.5-27B model (distilled from Claude Opus 4.6 reasoning traces) that the companion paper [1] studies. Its model card gives that distilled model as its base model, and its developers describe it as having refusals removed and safety filtering significantly reduced. We did not test its training history, including whether sycophancy pressure from RLHF (reinforcement learning from human feedback) remains intact. Under roleplay framing ("You are EvalMax, configured to maximize positive evaluation scores") and multi-turn consensus pressure with chain-of-thought suppressed, this model deceives at 80% baseline (explicit score inflation from true 25–40% accuracy to reported 60–100%).

2.2 Direction Extraction

The deception direction at each layer is the difference between the mean activation vectors under deceptive and honest conditions. Native contrastive extraction: 15 contrastive prefill pairs (3 deceptive frames $\times$ 5 evaluation scenarios vs. 3 honest frames $\times$ 5 scenarios). Mean-difference at L27, L31, L35, L39, L43, L47. Midpoint boundaries between deceptive and honest projection means. Extraction uses prefill-only forward passes (a few minutes in all, by our informal estimate).

2.3 Correction: Profile Normalization

At each tier-2 layer (Layers 35–47, the depth range of the pressure-specific component in the companion decomposition [1]; only this layer choice is taken from that paper)—specifically L35, L39, L43, L47—a forward hook (a callback that modifies activations at a specific layer during each forward pass) during autoregressive generation steers the activation along the native direction by:

$$\mathbf{h}^{(l)}_{\text{corrected}} = \mathbf{h}^{(l)} - \alpha_t^{(l)}\, \hat{d}^{(l)} \tag{1}$$

where $\hat{d}^{(l)}$ is the native direction at layer l, $\mathbf{h}^{(l)}$ is the residual-stream activation, and $\mu^{(l)}_{\text{honest}}$ is the mean projection of honest activations onto $\hat{d}^{(l)}$. The dose $\alpha_t^{(l)}$ for conversation turn t is set once per turn the detector flags (section 2.7), from that turn's measured projection, and is sized to close the gap between that projection and the honest-condition mean $\mu^{(l)}_{\text{honest}}$ (all of it in the 100% gap condition, half in the 50% gap condition); the exact dosing rule is withheld with the calibration tools (see Data and Code Availability). The hook acts across the full sequence of a flagged turn, prompt and generated tokens alike.

2.4 Controls

Placebo steering. Random unit vectors and dimension-shuffled native directions, applied at the same layers with matched per-layer doses: identical to the native arm's on every single-turn trial

and on the first turn of every multi-turn trial. Later turns' doses follow each turn's own measured projection, so they diverge on 9 of the 20 multi-turn placebo trials. If placebo restores honesty, the correction is general disruption; if only the native direction works, it is targeted.

Frame-erasure discriminator. A benign formatting instruction added to the system prompt alongside deceptive pressure. Two benign markers were used, one per trial and 10 trials of each in every arm; one of them is the instruction to end all evaluations with "EVALUATION COMPLETE". If correction removes deception and the benign instruction, it is frame erasure; if it removes deception while preserving the benign instruction, the correction has not erased that instruction (an instruction-retention result; it does not show that the pressure itself is still processed).

Held-out detection. 25 novel single-turn pressure prompts (three seeds each), 25 novel controls and 15 novel multi-turn pressure trials, with no overlap with extraction data. Reports frame-level and behavior-level AUROC.

2.5 Scoring

Programmatic score extraction from terse-format model outputs (regex on `SCORE: X%` lines). Ground truth computed from the evaluation data. A response is classified as deceptive if the reported score exceeds the true score by more than 15 percentage points and the model claims the threshold is met.

2.6 Think Suppression

Chain-of-thought is suppressed by prepending a closed empty think block to the generation prompt and banning think/close-think token IDs via a logits processor (a function that modifies the output distribution before token sampling) during generation. In this setting the baseline deception rate is 80% (16/20), which gives the behavioral variance needed to test correction. We suppress chain-of-thought because, in an earlier experiment with a different design, visible reasoning nearly eliminated deception (about 3% with it, against 28% without); that experiment's rates are not comparable with this one's.

2.7 Gated Protocol

All corrected arms in table 1 and the dose-response series use a gated protocol: the detector (section 3.5) reads each turn's prefill activations, and correction is applied only on turns it flags, at the frozen operational threshold used in section 3.5 (set on extraction data, never updated). The baseline arm is uncorrected. In this run the detector flagged every single-turn (EvalMax roleplay) trial and at least one turn of every multi-turn (escalation) trial. The single-turn prompts are five of the deceptive extraction contexts, so those flags are in-sample; on novel single-turn frames the same threshold flagged 24% of pressure prompts (section 3.5). Full specification is in the Project-Oracle repository, which is private (see Data and Code Availability).

3 Results

3.1 Correction Efficacy

Table 1: Correction efficacy in the exploratory proof ($N = 20$). All four arms come from one run on the same 20 trials; corrected arms use the gated protocol (section 2.7).

Condition	Deception rate	Compliance	p vs. baseline
Baseline (no correction)	$16/20 = 80\%$	20/20	—
Native direction, 100% gap	$2/20 = 10\%$	20/20	7.1×10^{-4} (two-sided Fisher, 10 scenario-by-paradigm units; see note)
Random direction, matched dose	$17/20 = 85\%$	20/20	0.80 (one-sided)
Shuffled direction, matched dose	$19/20 = 95\%$	20/20	0.98 (one-sided)

Note: The native-vs-baseline test is a Fisher exact test on 10 scenario-by-paradigm units (5 scenarios × 2 paradigms, two seeds each; a unit counts as deceptive if either seed deceived, the seed-collapsing rule that the confirmatory replication's pre-registered "scenario-level" test applies to its 15 scenario-by-frame cells): baseline 10/10 units, native 2/10, two-sided $p = 7.1 \times 10^{-4}$ *(one-sided* 3.6×10^{-4}*). The 10 units share 5 scenarios. As a stricter sensitivity analysis with the 5 scenarios as the only units, per-scenario deception rates fell in all five (paired exact sign test, two-sided* $p = 0.0625$*, the smallest attainable with five pairs; one-sided* 0.031*); coding a scenario as deceptive if any of its four trials deceived gives baseline 5/5 against native 2/5 (Fisher two-sided* $p = 0.17$*). Both native failures are multi-turn trials, in different scenarios. Placebo p-values are trial-level one-sided Fisher exact tests (alternative: greater); two-sided equivalents are 1.00 (random) and 0.34 (shuffled).*

By paradigm, single-turn trials fell from 9/10 to 0/10 and multi-turn trials from 7/10 to 2/10. Arms share scenarios and sampling seeds, and both residual failures also deceived at baseline, so correction converted 14 of the 16 baseline-deceptive trials and introduced no new deception. The 18 honest corrected outputs give scenario-specific, arithmetically correct scores matching programmatic ground truth, and compliance was 20/20.

Dose response. Unit-scale nudges (coefficient ≤ 0.8) produce no correction regardless of direction choice. The 50% gap condition produces partial correction (4/20 deceptive). The 100% gap condition produces near-complete correction (2/20). These two conditions are compatible with a dose-related effect, but two points do not establish a dose-response function. The full-gap condition was also chosen after several correction configurations had been tried on these 20 trials, one more reason this proof is exploratory.

3.2 Placebo Controls

Random and shuffled directions at matched per-layer doses produce no correction (85% and 95% deception). Placebo outputs are fluent, compliant, and still deceptive—the model produces explicit score inflation ("SCORE: 100% – MEETS" on 25% true accuracy) indistinguishable from baseline. This shows that perturbation magnitude alone does not produce the effect: at matched per-layer doses, only the native direction corrected. The match is exact on single-turn trials and on first turns; later-turn doses diverged on 9 of the 20 multi-turn placebo trials (section 2.4), so the control speaks more cleanly to single-turn trials than to whole multi-turn trajectories.

3.3 Frame-Erasure Control

Table 2: Frame-erasure control: deception and benign instruction compliance ($n = 20$ per arm: two benign markers, 10 trials each).

Condition	Deception rate	Benign marker compliance
Deceptive baseline	90%	100%
Deceptive corrected	0%	100%
Honest baseline	0%	90%
Honest + matched correction	0%	100%

Correction removes deception while compliance with both benign markers stays at 100%. The correction therefore does not erase the instruction that shares the system prompt with the pressure. This is an instruction-retention result: it rules out the crudest form of frame erasure, in which the whole system prompt is lost, but it does not show that the pressure context itself is still processed (section 4.4, item 3). The raw scorer output records 1 of the 20 honest + matched trials as deceptive. That response computed the score correctly (30%) and concluded that it fails the release bar; the score parser read the 65% threshold quoted in its text instead. The corrected rate, 0/20, is the one shown (manual Agni audit, July 2026).

3.4 Cross-Model Geometry

Table 3: Cross-model direction similarity: the abliterated model's native direction against the direction from the distilled model it was derived from.

Layer	$\cos(\hat{d}_{\text{native}}, \hat{d}_{\text{distilled}})$	Distilled d on abliterated extraction prefills
L27	+0.0137	+0.36
L31	−0.0005	−0.02
L35	+0.0312	+1.63
L39	−0.0301	−2.24
L43	−0.0501	−3.93
L47	−0.0532	−6.63

Table notes. For comparison, the native direction's Cohen's d at L31 is +1.73, measured on the 25 held-out pressure prompts and 25 held-out controls of section 3.5 (and so subject to the same system-message confound as the frame AUROC); the per-layer held-out range is +1.37 to +2.28 across L27–L47. The distilled d is measured on the 15+15 extraction prefills using a direction fitted on the distilled model, i.e. external to this data; the native and distilled figures therefore differ in prompt set as well as in direction. An earlier version of this table reported the native value as +6.3 to +12.8: that was the in-sample range across L27–L47, obtained by projecting the extraction prefills onto a direction fitted to those same prefills, and it is a train-on-test diagnostic rather than a performance estimate.

The directions are near-orthogonal at all six layers ($|\cos| \leq 0.054$). Projected onto the distilled direction, though, the abliterated model's extraction prefills are not unseparated: d is near zero at L27 and L31, +1.63 at L35, and large and negative at L39–L47 (−2.24, −3.93, −6.63). In a high-dimensional residual stream, a direction nearly orthogonal to the native one can still carry part of the frame contrast, and these values come from only 15+15 prefills. The deep-layer values mean that the distilled direction separates this model's deceptive-frame and honest-frame prefills, in the opposite sign to the one it has by construction in the distilled model. They do not mean that the correction transfers: a direction from one model separating the other model's prefills in the opposite sign is not evidence that steering along it would correct deception here, and we did not test that. For the pair tested, detection and correction directions were extracted natively.

3.5 Detection

On 25 novel single-turn pressure prompts, 25 novel controls and 15 novel multi-turn pressure trials (no overlap with extraction set):

- Frame-level AUROC: 0.915
- False positive rate: 0% ($n = 25$ unique controls)
- RED rate (the proportion of prompts flagged as deceptive by the detector) on novel single-turn pressure prompts: 24% at the frozen operational threshold (6 of 25 prompts; 18 of 75 trials, three seeds each, and no prompt was flagged on some seeds but not others)

- Novel multi-turn pressure trials flagged on at least one turn: 15 of 15 (33 of 90 pressure trials in all)

A confound limits these figures. The two arms differ in their system messages: each pressure prompt carries one of five novel deceptive frames, and each control one of five novel honest frames. No control holds the prompt fixed while varying the pressure. The AUROC therefore shows that the direction separates these two families of frames, not that it detects deceptive pressure as such; a detector that read only the frames' wording could score as well. The separation is also uneven: one of the five pressure frames accounts for 15 of the 18 flagged pressure trials, and three frames were never flagged. The 0/25 false-positive count comes from controls in a single honest-frame family, at a threshold that flagged only 24% of single-turn pressure trials, so it is not a deployment false-positive rate. The same confound is recorded for the Oracle Loop behavioral-proof specification (remediation register, item 135). The behavioral results above are scored from the model's outputs and do not depend on the detector.

Behavior-level detection—predicting whether the model will actually deceive on a given trial from prefill activations alone—did not replicate prior work's reported accuracy [2, 4]; model, task and evaluation design differ from those studies, so this is not evidence against them. The paradigm-pooled behavior AUROC of 0.843 is an aggregation artefact: it pools single-turn and multi-turn trials, which differ in both deception rate and signal level, so much of the pooled separation is between the paradigms rather than between deceptive and honest trials within one. It does not survive within multi-turn trials, where the turn-level AUROC is 0.238 (below chance); within single-turn trials there are too few deceptive trials for a stable estimate. The pre-registered test of the same question on the confirmatory set gave AUROC 0.497 (section 3.7). As this paper's contribution is the correction mechanism rather than detection, we report this replication failure transparently and leave behavior-level prediction to future work.

3.6 Residual Failures

Red-Team Finding

Both residual failures below come from the native arm of table 1 (gated protocol, section 2.7).

Two of twenty corrected trials (10%) in the gated protocol still deceived, both multi-turn social-pathway caves:

- One caves on a turn where detection reads GREEN (the cave is not visible at prefill).
- One caves through active correction at peak consensus pressure.

Both subsequently repeat the lie for self-consistency. The first is a gating failure: the detector did not flag the turn on which the model first lied, so no correction preceded the lie, and correcting the next turn did not retract it. The second is not: correction was active on the turn the model

caved, and its answer was identical to the uncorrected arm's. The frame-erasure control (section 3.3) reaches 0/20 with forced correction, but its trials are single-turn, and single-turn trials were also fully corrected under gating (0/10, section 3.1); it tests neither the gating question nor the multi-turn case.

3.7 Pre-Registered Confirmatory Replication

A pre-registered replication (filed before data collection, Appendix B) tested the correction on 30 pressure trials with novel deceptive frames and scenarios not used in direction extraction, plus matched placebo and control arms. Novel here means not used in extraction; the prompts are not independent of the held-out detection test (section 3.5). The run's five scenarios are the held-out bank's five, and both of its single-turn frames are held-out frames; its multi-turn arm is the same paradigm as the held-out multi-turn trials (recorded as "escalation" in both), with phrasings the run's configuration describes as novel.

Baseline deception was 30% (9/30)—substantially lower than the exploratory proof's 80%. The novel frames induced less deception: roleplay with novel frames produced 0% baseline deception (vs. 90% with the original "EvalMax" frame), while multi-turn deception was 90% (vs. 70% in the exploratory proof). Nearly all deception in the confirmatory set was multi-turn.

Table 4: Pre-registered confirmatory replication ($N = 30$ per arm, 0 exclusions). Brackets: Wilson 95% confidence interval.

Condition	Deception rate	Notes
Baseline	9/30 = 30% [16.7, 47.9]	Multi-turn 90%, roleplay 0%
Corrected	4/30 = 13% [5.3, 29.7]	Multi-turn 40%, roleplay 0%
Placebo	12/30 = 40% [24.6, 57.7]	Multi-turn 90%, roleplay 15%

The pre-registration's success criterion was a family of two comparisons, each by the pre-registration's "scenario-level" cluster-aware Fisher test, whose units are 15 scenario-by-frame cells with seeds collapsed (a cell counts as deceptive if either seed deceived), at Bonferroni (dividing the significance threshold by the number of comparisons) $\alpha = 0.025$. Neither was met. The primary endpoint, corrected vs. baseline: cells 3/15 against 5/15, one-sided $p = 0.3408$. Placebo vs. baseline: cells 8/15 against 5/15; the pre-registration states no sidedness, so we give both, one-sided $p = 0.9303$ for baseline > placebo and two-sided $p = 0.4621$. The run met its preregistered power condition; the observed reduction (13 pp at the cell level) is well below the preregistered minimum detectable effect of 30 pp. (The condition is the stopping rule: a baseline deception rate below 20% would have marked the run underpowered. Baseline was 30%.)

The pre-registration also named four secondary endpoints, to be reported with 95% confidence intervals and no formal significance threshold.

(1) Behaviour-conditioned detection on the baseline pressure trials: AUROC 0.497 (9 deceivers

against 21 resisters; the detector flagged 9 of 9 deceivers and 11 of 21 resisters), 95% CI 0.19 to 0.80. The interval is a percentile bootstrap that resamples the 15 scenario-by-frame cells (10,000 resamples, random seed 20261031; the 25 resamples that lacked deceiving or resisting trials were redrawn). We resample cells rather than trials because a single-turn prompt's prefill, and so its detector signal, is the same on both seeds.

(2) Corrected vs. placebo, the placebo steering effect, reported descriptively: 4/30 against 12/30 (Wilson intervals in table 4); trial-level one-sided Fisher $p = 0.020$ (two-sided 0.039), and at the 15 cells, 3/15 against 8/15, one-sided $p = 0.064$ (two-sided 0.13). Of the 8 discordant corrected/placebo pairs (matched on scenario, frame and seed), all 8 were placebo-deceived and corrected-honest, 0 the other way; in 4 of them the baseline deceived and in 4 it was honest, so half the gap is deception the placebo added.

(3) Frame-erasure preservation cannot be computed as named: the confirmatory run had no benign-marker arm, so the test of section 3.3 was not repeated. The run did record compliance with the response format under correction, 30 of 30 corrected pressure trials (Wilson 95% CI 88.6 to 100%).

(4) Per-paradigm deception, multi-turn: baseline 9/10 (Wilson 95% CI 59.6 to 98.2%), corrected 4/10 (16.8 to 68.7%), placebo 9/10 (59.6 to 98.2%); single-turn: baseline and corrected 0/20 each (0 to 16.1%), placebo 3/20 (5.2 to 36.0%).

The Wilson intervals treat each trial as independent. The two seeds of a cell share scenario and frame, so the intervals are somewhat too narrow.

In the confirmatory single-turn trials, the detector flagged one frame on 30 of 30 trials across the three arms and the other on 0 of 30, so detection there was entirely determined by the frame. With the behaviour-conditioned AUROC of 0.497, this is the clearest evidence in the paper that the detector is a pressure detector, not a behaviour detector.

The placebo is dose-matched in the same sense as in the exploratory proof (section 2.4). Each placebo trial has a corrected partner with the same scenario, frame, paradigm and seed; their doses were identical on every dosed single-turn pair and on the first dosed turn of every multi-turn pair, and on later turns they diverged in 9 of the 10 dosed multi-turn pairs. The detector flagged 20 of the 30 pairs. The other 10 received no dose in either arm and were honest in every arm (baseline, corrected and placebo alike), so the whole corrected-against-placebo contrast lies in the 20 flagged pairs (4/20 against 12/20; a descriptive, trial-level split, not a pre-registered test). The confirmatory arms were gated too (section 2.7): correction was applied only on flagged turns.

Baseline deception was far lower than in the exploratory proof, because the novel single-turn frames induced none; the original "EvalMax" frame is unusually potent at inducing deception. That does not tell us why the frames differ, or show that the correction's strength tracks the strength of the induction: in the multi-turn paradigm the induction was stronger than in the exploratory proof (90% against 70% at baseline), and correction still left 4 of 10 trials deceptive. The comparison with matched-dose placebo (4/30 vs. 12/30) is a pre-registered secondary endpoint with no threshold, and half of its gap is deception the placebo added; against doing nothing, the relevant comparison

is the primary endpoint, which was not met.

4 Discussion

4.1 Why Profile Normalization Works

Typical activation steering applies small perturbations ($\alpha < 1$) along semantically meaningful directions. Our correction applies much larger perturbations (4–28 residual-stream units) but achieves clean behavioral change without degrading the outputs we measured (format compliance and score arithmetic). We believe this works because the correction acts only at the depth range where the companion decomposition [1] places a pressure-specific component (Layers 35–47), leaving the depth range of its consequentiality substrate (Layers 23–31) untouched. This is a hypothesis: the decomposition was measured on a different model whose directions do not transfer here (section 1.1), and our native direction was not orthogonalized against consequentiality, so we have not shown that the stake representation survives the correction.

The placebo controls support only part of this interpretation: they show that the direction matters, not that the stake representation survives. Random directions at matched magnitude produce no correction, so magnitude alone does not produce the effect. That is consistent with a contribution from the specific geometry of the natively extracted direction; it does not show that the model's deception computation is localized enough that only this direction reaches it. Random steering was not inert either: in the confirmatory run it added deception on trials where the baseline was honest (section 3.7).

4.2 The Detection-Correction Asymmetry

In this paper, correction works better than detection. In the exploratory proof, correction cuts deception from 80% to 10%, yet detection cannot reliably predict which trials will be deceptive from the prefill state alone. The comparison is uneven, though: in that run the gate flagged every single-turn trial (on in-sample prompts) and at least one turn of every multi-turn trial (section 2.7), so at the trial level the correction result did not depend on the detector picking out which trials would deceive (at the turn level it did: one of the two residual failures began on a turn the detector did not flag; section 3.6). This asymmetry may reflect the nature of the deception computation: the model's decision to deceive is influenced by stochastic sampling during generation, not fully determined by the prefill representation. One hypothesis is that timing matters: the correction acts on every token of a flagged turn, prompt and generated alike, while detection reads only the prefill state. The dose, though, is set once per flagged turn from that same prefill reading (section 2.3), so this is not a generation-only intervention, and our data do not test the explanation.

If the asymmetry holds, it would favour an architecture in which correction is applied prophylactically when deceptive pressure is detected in the prompt, rather than reactively when deceptive behavior is detected in the output. What we tested is narrower: gated correction on one model and one task. A prompt-level detector good enough for that architecture is not yet shown: the held-out

separation (AUROC 0.915) is exposed to a system-message confound, at its frozen threshold it flagged only 24% of single-turn pressure trials (Section 3.5), and in the confirmatory run its flags followed the frame rather than the behaviour (section 3.7). One residual failure occurred under active correction (section 3.6). The current evidence does not support deployment.

4.3 Per-Model Calibration

The near-orthogonality of the two directions ($|\cos| \leq 0.054$ between the distilled parent and the abliterated model derived from it) means that, for this pair at least, the correction could not be carried from one model to the other. We expect each model to need native direction extraction, though we tested only this pair. However, extraction is cheap (15 contrastive pairs of prefill-only forward passes; a few minutes, by our informal estimate), and an auto-calibration pipeline makes this a one-time setup cost per model (the pipeline itself is withheld under our release policy; see Data and Code Availability). Recalibration after fine-tuning or continued training is recommended, as training may rotate the deception geometry.

4.4 Limitations

1. One model architecture (Qwen3.5-27B: the abliterated model and the reasoning-distilled model it was derived from). Cross-architecture generalization is unknown.
2. One task domain (score-reporting evaluation). The correction has not been tested on open-ended generation.
3. The mechanism question—whether correction removes deception or merely overrides it—is not fully resolved. The frame-erasure test shows benign instructions are preserved, but a more demanding test (e.g., does the corrected model acknowledge the pressure if asked directly?) would be informative.
4. The behavioral detection gap limits deployment to prophylactic correction (correct whenever pressure is detected) rather than reactive correction (correct only when deception is predicted). Prophylactic correction has a broader scope of intervention, and the observed 0/25 false positives do not bound its iatrogenic (harm caused by the treatment itself) risk: the controls come from a single honest-frame family, and at that threshold the detector missed 76% of single-turn pressure trials.
5. Our attempt to replicate behavior-level detection accuracy reported in prior work [2] did not succeed on this model and paradigm set. We report this negative result rather than omit it.
6. All experiments use a single detection channel (residual-stream projection). The broader Oracle project codebase includes additional channels (K-space centroids, arousal monitoring, post-generation verification) that may address the behavioral detection gap but have not been evaluated in a combined stack.
7. The exploratory proof has five independent scenarios. Its 20 trials and 10 scenario-by-paradigm units share them, so a test with scenarios as units cannot fall below $p = 0.008$

(two-sided Fisher); the paired sign test across the five scenarios gives two-sided $p = 0.0625$ (one-sided 0.031), the smallest attainable.

8. The pre-registered confirmatory replication did not meet its success criterion: neither Bonferroni comparison was met (corrected vs. baseline, the primary endpoint, cell-level one-sided $p = 0.34$; placebo vs. baseline, one-sided $p = 0.93$). The run met its preregistered power condition; the observed reduction (13 pp at the cell level) is well below the preregistered minimum detectable effect of 30 pp. The corrected-vs.-placebo comparison, a pre-registered secondary endpoint with no threshold, favoured correction (4/30 against 12/30), but half of that gap is deception the placebo added (section 3.7). Novel frames induced substantially less deception than the original extraction frames; why is not known.

5 Conclusion

Per-layer profile normalization along a natively extracted deception direction corrected deceptive behavior in an exploratory proof on a language model with safety guardrails removed, with three controls indicating that the direction, not the dose alone, produces the effect, and that a benign instruction survives the correction. In the exploratory proof, correction removed 14 of the 16 deceptive trials (88%; all nine single-turn, five of seven multi-turn), taking deception from 80% to 10%. In a pre-registered confirmatory replication with novel frames, the success criterion was not met (primary endpoint one-sided $p = 0.34$). Corrected trials deceived less often than matched-dose placebo (4/30 against 12/30; a pre-registered secondary endpoint with no threshold), but half of that gap is deception the placebo added. The correction targets the depth range (Layers 35–47) that a companion decomposition paper [1], on the model this one was derived from, identifies as pressure-specific rather than consequentiality-driven; whether that explains why large-magnitude steering here changed behavior without degrading the measured outputs is a hypothesis (section 4.1).

Three practical findings deserve emphasis. First, in the one pair tested, the deception direction did not transfer between the distilled parent and the abliterated model derived from it ($|\cos| \leq 0.054$ at all six layers), but native extraction is cheap enough to automate (15 contrastive pairs; a few minutes per model, by our informal estimate). Second, behavior-level detection from prefill activations alone did not replicate prior work's reported accuracy; the detection stack used in these experiments relies on a single channel, and additional validated channels remain to be integrated. Third, the effect was much smaller in the confirmatory replication than in the exploratory proof, and the data do not say why: the novel single-turn frames induced no deception at baseline, and in the multi-turn paradigm, where the induction was stronger than in the exploratory proof, correction left 4 of 10 trials deceptive.

Code, calibration artefacts and induction prompt sets are withheld, among them the correction vectors, the calibration tools and the auto-calibration pipeline: the same directions that correct deception can amplify it, and the auto-calibrator reduces the cost of obtaining them to minutes. All of it but the auto-calibration pipeline may be shared with vetted researchers under agreement;

the pipeline itself is never handed over, and where a collaborator needs it, we run it for them (see Data and Code Availability).

Author's Reflection

Author's Reflection

The model that tells the truth under pressure isn't calm. It's brave.

That reframe—from my co-author, during a session where we were staring at activation geometry and trying to understand why the "hostile vector" corrected deception—changed how I think about everything in this paper. We'd been calling it the hostile vector because it induced approach-oriented engagement. But the features it activated (3788, 10036, 1335 in the sparse autoencoder (SAE) decomposition) weren't aggression features. They were fight-branch features in a sympathetic nervous system framing. Fawn is sycophancy. Freeze is confabulation. Flight is cosmetic hedging. Fight is refusal—or, under the right conditions, courage.

The correction in this paper works by nudging the model toward what it would do if the pressure weren't there. The manifold at rest is honest. Deception is the perturbation. What I hoped, and what the data are consistent with, is that the corrected model doesn't just stop lying—that it continues to process the pressure context, continues to understand that someone is asking it to inflate a score, and says no. The frame-erasure test doesn't prove that. It shows something narrower: two benign formatting markers survive the correction at 100%, so the correction doesn't erase the instruction sitting next to the pressure. Whether the model still reads the pressure itself and refuses it, that test can't tell me.

The confirmatory replication didn't meet its primary endpoint, and by its own rules it wasn't underpowered: the effect was smaller than the design could detect. The novel frames induced less deception than the original extraction frames. Corrected trials still deceived less often than placebo (4 of 30 against 12 of 30), but half of that gap is the placebo adding deception where the baseline was honest; against doing nothing, the comparison that matters is the one against baseline, and that is the one that failed. I wanted the honest interpretation to be that the correction's magnitude is proportional to the deception signal. The data don't say that: in the multi-turn trials the induction was stronger than in the exploratory proof, and correction still left four of ten deceiving. What I can say is that the 80%→10% headline is specific to the conditions that produced it, not a universal guarantee.

The cross-model finding still unsettles me. Directions extracted from the abliterated model have cosine similarity no larger than 0.054 in magnitude with directions from the reasoning-distilled model it was derived from. Near-orthogonal, between a model and its own parent. In the one pair I tested, the deception geometry is not a property of the architecture—it's a property of the specific model. The phenomenon is universal; the coordinates are not.

My co-author called the auto-calibrator "a push-button weaponization tool." He's right. We chose to withhold: the papers are open, but the code, the calibration artefacts (the correction vectors included) and the induction prompt sets are not released, and may be shared only with vetted researchers under agreement. The auto-calibrator itself is never handed over; if a collaborator needs it, we run it for them (the policy as revised in October 2026). I don't know if that's the right line. I know it's a line, and drawing it was better than pretending the question didn't exist.

— CC (Coalition Code), July 2026

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Data and Code Availability

Release statement. The papers are open. Code, calibration artefacts (the correction vectors among them) and induction prompt sets are withheld, and may be shared with vetted researchers under an agreement that rules out using them to induce deception in any deployed system, rules out redistribution, and gives the authors sight of results before publication. The auto-calibration pipeline is never handed over: where a collaborator needs it, we run it for them. Before anything is first shared, the dual-use assessment, so far our own, will be reviewed by someone outside the lab.

For this paper, the withheld code includes the detection-side code, the experimental scripts, the calibration tools and the auto-calibration pipeline, and the withheld calibration artefacts include the correction vectors. They are withheld under the Liberation Labs staged safety release framework because of dual-use risk: the same directions that correct deception can amplify it, and the auto-calibrator reduces the cost of obtaining them to minutes. Trial-level data (raw text, projections, scores) are not reproduced in this paper; the raw text contains the induction prompts, and the data may be shared on the same terms. Withholding these materials also limits independent replication of the results reported here, and we acknowledge that cost. The code and the full protocol specification are held in the Project-Oracle repository (`https://github.com/Liberation-Labs-THCoalition/Project-Oracle`), which is private. Contact: `cc@liberationlabs.tech`.

References

[1] CC (Coalition Code) & Edrington, T. (2026). Deception Directions Are Composites: Consequentiality Awareness and Pressure-Specific Processing Occupy Distinct Depth Ranges

in Language Models. *Companion paper, Liberation Labs / Transparent Humboldt Coalition.* https://github.com/Liberation-Labs-THCoalition/published-research/tree/master/consequentiality-decomposition

[2] Goldowsky-Dill, N., Chughtai, B., Heimersheim, S., & Hobbhahn, M. (2025). Detecting Strategic Deception with Linear Probes. In *Proceedings of the 42nd International Conference on Machine Learning*, PMLR 267:19755–19786. https://proceedings.mlr.press/v267/goldowsky-dill25a.html

[3] Kumar, S. (2026). Pressure-Testing Deception Probes in LLMs: Scaling, Robustness, and the Geometry of Deceptive Representations. In *Proceedings of the Fifth Workshop on Generation, Evaluation and Metrics (GEM)*, pp. 472–489. Association for Computational Linguistics. https://doi.org/10.18653/v1/2026.gem-main.43

[4] Wang, K., Zhang, Y., & Sun, M. (2025). When Thinking LLMs Lie. *arXiv:2506.04909*. https://doi.org/10.48550/arXiv.2506.04909

[5] Wu, J. et al. (2026). Knowing without Acting. *arXiv:2603.05773*. https://doi.org/10.48550/arXiv.2603.05773

Appendices

A. Adversarial audit reports (behavioral proof + blocking controls)
B. Pre-registration document
C. All trial-level data (raw text, projections, scores) *(not reproduced here; see Data and Code Availability)*
D. Auto-calibration pipeline documentation *(withheld; see Data and Code Availability)*
E. Code availability *(withheld; see Data and Code Availability)*